Effects of Pre-Processing Parameters on Stretch Reflex EMG: A Comparative Analysis

Connor A. Spinelli, Rebecca C. Nikonowicz, Dr. Fabrizio Sergi
Human Robotics (HuRo) Laboratory, University of Delaware; Delaware INBRE, University of Delaware contact: cspin@udel.edu, fabs@udel.edu

Introduction

- Human motor control involves task-dependent motor responses, modulated by several supraspinal motor pathways
- The **long latency response (LLR)**, a stretch reflex, triggered by a sudden muscle stretch, is a crucial part of the task-dependent motor response [1]
- We use **surface electromyography (sEMG)** to accurately measure muscle activity during a stretch evoked motor response

Importance of Pre-processing

- Raw sEMG signal features contain noise and artifacts that can be removed with **pre-processing** which isolates clean physiological signal [2]
- Standard EMG pre-processing pipelines include 3-steps: bandpass filtering, rectification, and linear enveloping [2,3]

Problem

- sEMG signal interpretation is highly dependent on pre-processing, but there are no universally agreed-upon pipelines for sEMG in stretch reflex studies [2]
- Lack of standardization raises concern that conclusions may be influenced by filter choice

Objective

- Verify the robustness of our findings from a previously collected experiment, ensuring the results aren't influenced by our pre-processing methods

Project goal

Investigate our pre-processing approach by examining how filtering parameters influence the statistical significance of LLR amplitudes across different task instructions.

Methods

Experimental Design

- 10 participants, in supine position with hand strapped into Dual Motor StretchWrist (DMSW) Robot [3]
- DMSW (**Fig. 2**) uses two piezoelectric ultrasonic motors to apply wrist joint extension perturbations for each of the three instructions: "Yield" (Y) to perturbation, "Resist" (R) perturbation, and a "Yield (Slow)" (YS) perturbation [3]

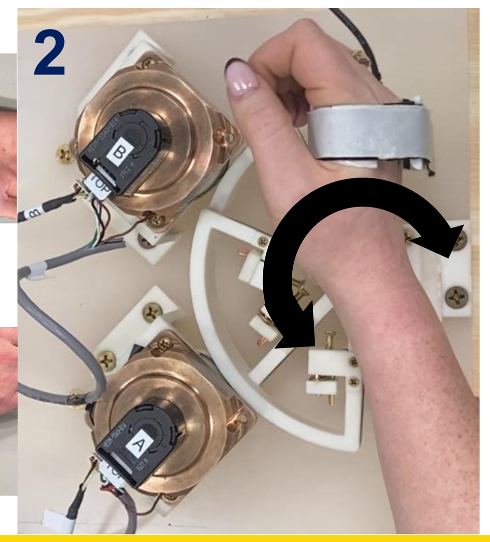
EMG Data Acquisition

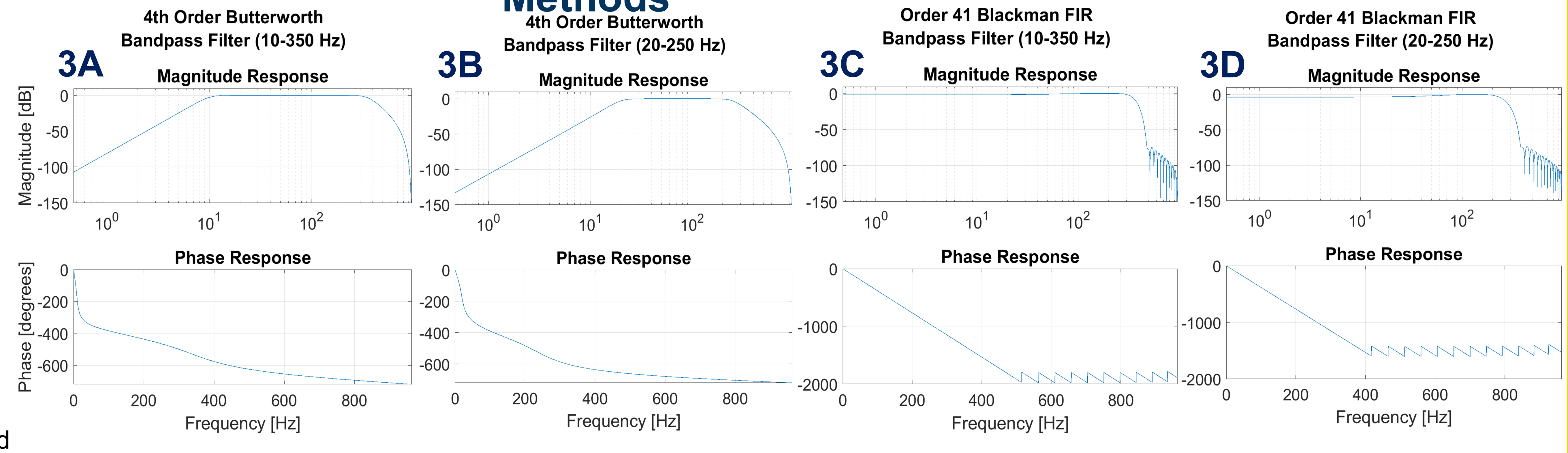
- Surface electrodes (Delsys Trigno Duo, Natrik, MA) sampled at 1926 Hz placed on the belly of the Flexor Carpi Radialis (FCR, **Fig. 1A**) and the Extensor Carpi Ulnaris (ECU, **Fig. 1B**) of the right forearm to measure muscle activity [3]

1A

1B

2





Pre-processing Techniques

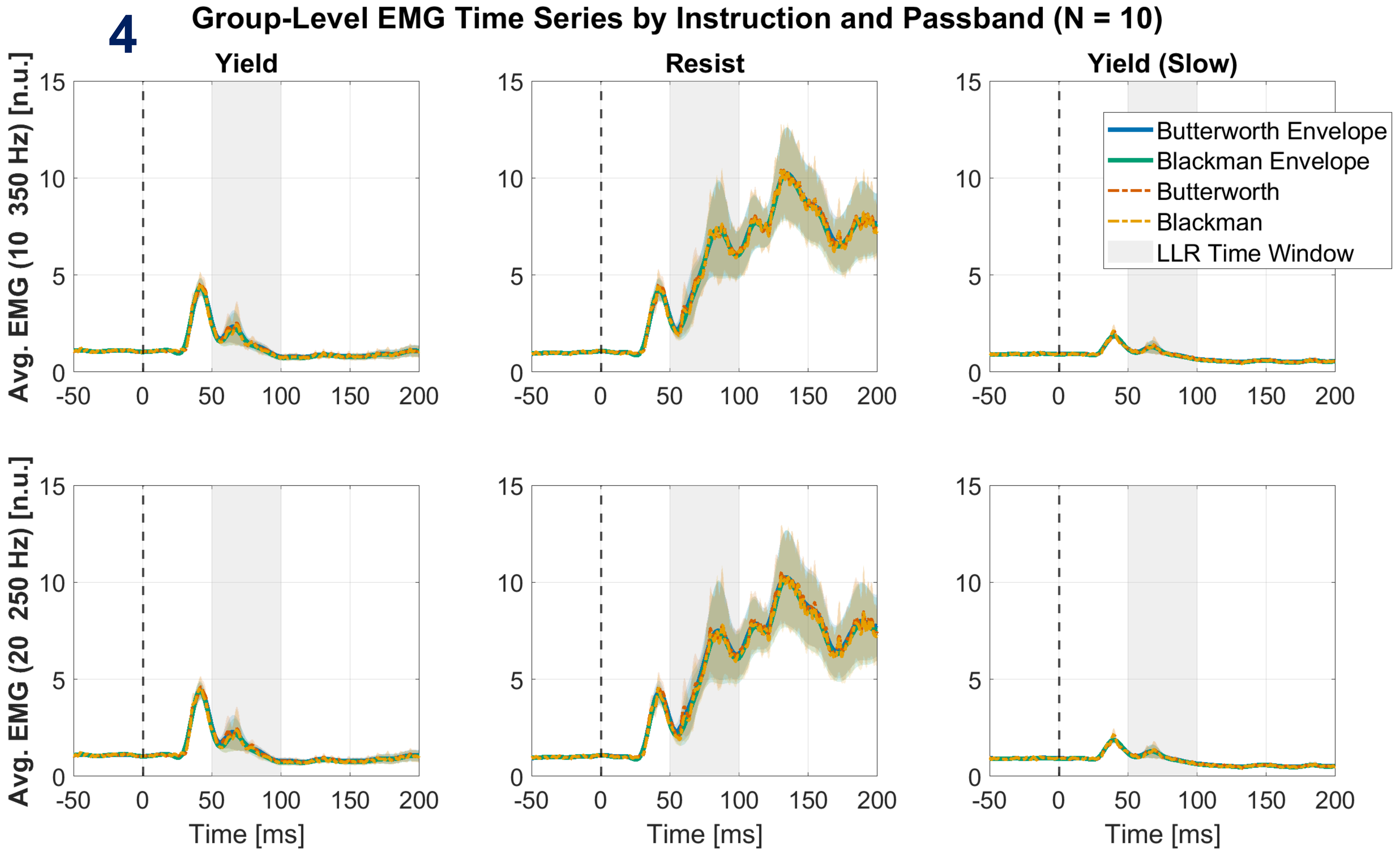
- 4 distinct pre-processing pipelines were applied to the FCR sEMG signal, each with a 10-350 Hz (**Fig. 3A and 3C**) passband and a 20-250 Hz (**Fig. 3B and 3D**) passband, generating 8 total filters
  - **4th Order Butterworth Bandpass Filter (Fig. 3A and 3B) + Rectification**
    - Isolates key EMG frequencies from noise, but its *non-linear phase shift* slightly alters the signal's timing
  - **4th Order Butterworth Bandpass Filter + Rectification + Linear Envelope**
    - The linear envelope adds a smooth envelope from the filtered signal, showing the overall intensity of the muscle contraction over time
  - **Order 41 Blackman FIR Bandpass Filter (Fig. 3C and 3D) + Rectification**
    - Uses a *linear phase* to preserve the exact timing of the muscle signal without introducing waveform distortion
    - Produces a wide transition band for cutoff frequencies
  - **Order 41 Blackman FIR Bandpass Filter + Rectification + Linear Envelope**
    - The linear envelope adds a smooth intensity profile of the contraction while avoiding the initial phase distortion common in other methods

Statistical Analysis

- Mixed Model Analysis in JMP with parameters:
  - *Fixed effects* of Instruction, Filter; with Interaction Instruction*Filter
  - *Random effects* of Subject; with Interactions Subject*Filter and Subject*Instruction
- Post-hoc Tukey HSD

Results

4



sEMG timeseries

- **Figure 4** is the average sEMG timeseries across 10 participants within each instruction
  - Clear distinction of LLR and then voluntary contraction (after LLR) when resisting the perturbation
  - sEMG timeseries resulting from the 4 different pre-processing pipelines with 10-350 Hz passband (top row) and a 20-250 Hz passband (bottom row)

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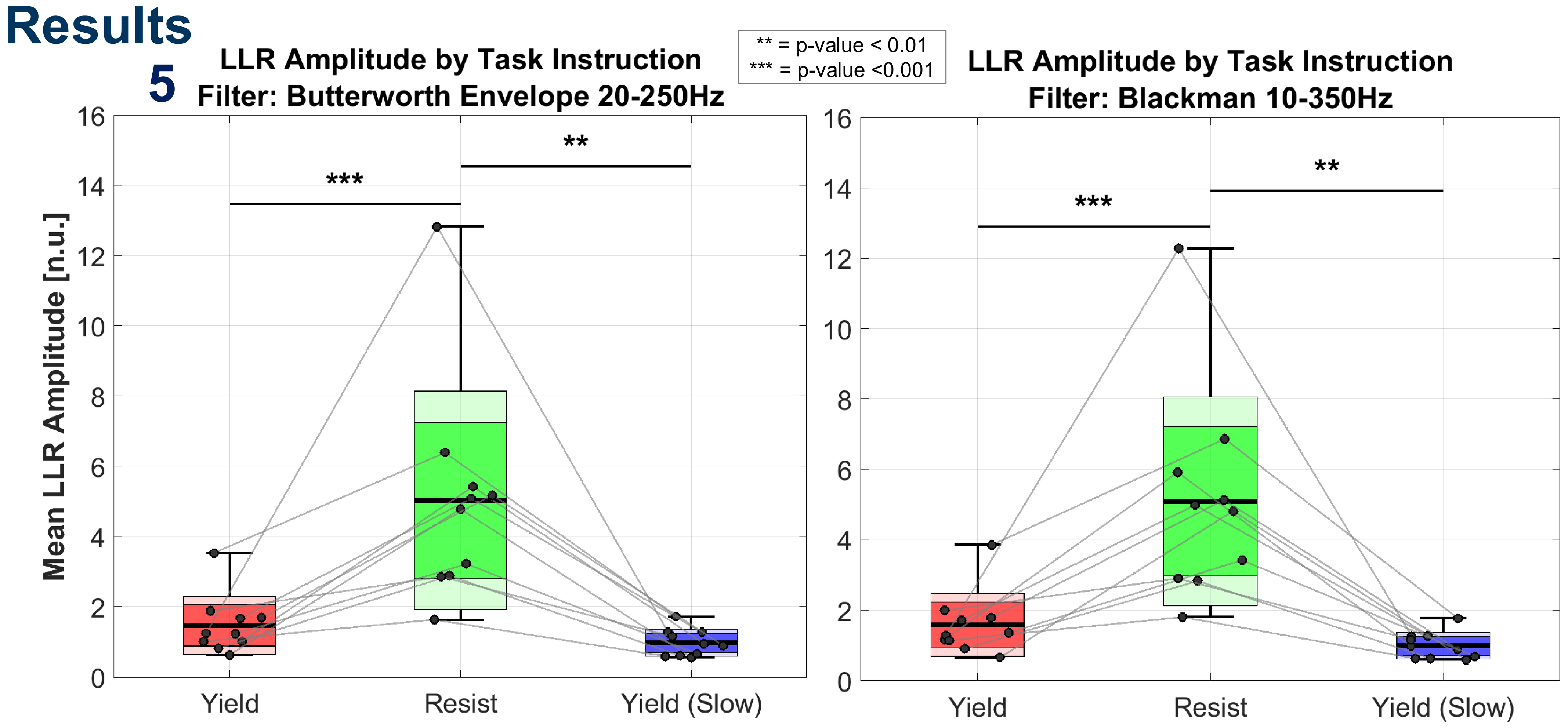

Statistical Analysis Results of LLR amplitude

- Results show significance in the Instruction and Filter Fixed Effects due to the difference of Resist, but no significance in the interaction of Instruction*Filter
  - *Resist vs. Yield* and *Resist vs. Yield (Slow)* showed a significant difference for all filters, while *Yield vs Yield (Slow)* showed no significant difference for all filters
  - *Resist vs. Resist, Yield vs. Yield,* and *Yield (Slow) vs. Yield (Slow)* all showed no significant difference for all filters
- **Figure 5** shows average LLR amplitudes for each instruction condition across all participants
  - Significant differences shown for *Resist vs Yield (Slow)* and *Resist vs. Yield* with p-values

Conclusions

- Only noticeable differences in sEMG timeseries include higher frequency amplitudes without an envelope
- Parameters can improve time-series wave-form to be more visibly interpretable
- Parameters have **no significant effect** on overall statistical analysis of amplitude
- Results are **consistent** with our standard approach to pre-processing, confirming that the conclusions are robust and not the result of pre-processing choice
- Results suggest that while sEMG filtering is important, the specific techniques chosen may not critically affect the significance of results in similar studies

References

[1] Kurtzer, et al.,. *Journal of Neurophysiology*, 100(4), 2008

[2] De Luca, Delsys Inc., Fundamental Concepts in EMG Signal Acquisition, 2003

[3] R. C. Nikonowicz and F. Sergi, *10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, 2024, pp. 1283-1288

Acknowledgments

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